

Prototype OTAs are held by Military Departments and the Defense Innovation Unit, AFWERX / SpaceWERX, Army Applications Lab and Naval Rapid Capabilities Office. When a prototype is successfully completed, the law explicitly allows a follow-on Production OTA without further competition, provided:

- The Prototype OTA was competitively awarded, and
- The Prototype OTA notice stated that follow-on production was possible. This is sometimes called the “prototype to production bridge.”

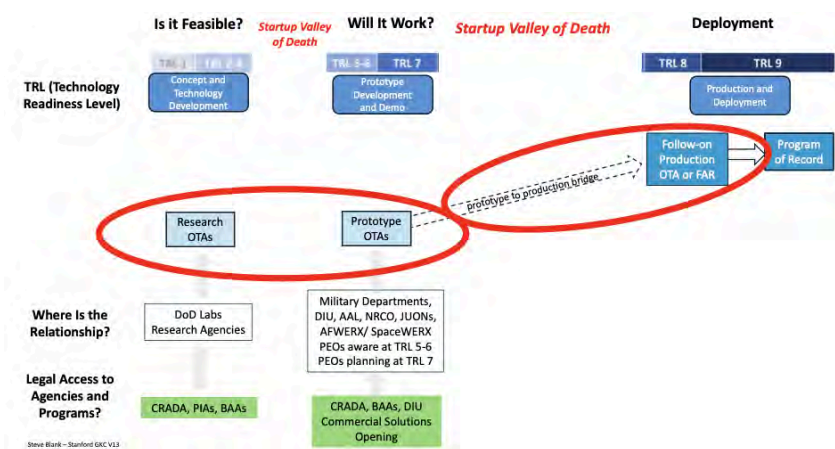
For startups the good news is that under OTAs Intellectual Property (IP) rights are fully negotiable. In general, companies retain ownership of IP assets created during the effort. The DoW usually licenses certain rights to use these assets with the agreed terms and conditions.

Many startups turn their SBIR Phase 2 awards into Prototype OTAs by using OTA Consortia. See the OTA Consortia section and Appendix B.

Production OTAs

Each Military Department (Army, Navy, Air Force) has Service Acquisition Executives (SAEs) who hold OTA authority. They typically delegate OTA authority down to Portfolio Acquisition Executives (PAEs) and then to their Capability Executive Offices (CPEs), contracting centers, or specialized offices. See the DoW DARPA [here](#).¹

In addition, the Army Contracting Command (ACC) has OTA-capable offices like ACC-New Jersey and ACC-Orlando. The Navy’s OTA execution often flows through NAVSEA or the ONR (Office of Naval Research). The Air Force Life Cycle Management Center (AFLCMC) executes AFWERX/AFOTEC OTAs.



R&D Agreements - CRADAs (Cooperative R&D Agreements)

CRADAs allow companies to work with DoW labs without procurement contracts. A DoW Laboratory can contribute people, services, facilities, equipment, intellectual property, data, and other resources, but not money. CRADAs don’t have funding attached to them, they can be a

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<https://acquisitioninnovation.darpa.mil/docs/Guides/Guide%20to%20Research%20Other%20Transactions%2009132023.pdf>

great way for startups to jointly pursue R&D with potential DoW customers and better understand end user problems, people, processes and requirements.

Successful Defense tech companies (Anduril, Saronic, Capella et al) all have had multiple CRADAs to engage in the detailed conversations needed to learn warfighter/customer requirements, and what to build. This typically occurred when they had high-fidelity prototypes. If companies use CRADA effectively, they should result in follow-on contract dollars.

Some DoW organizations/agencies can use a CRADA to get your people and your company cleared before you have a formal contract. One of the other benefits of a CRADA is that under a CRADA, a startup can access DoW test ranges. See Appendix C for a list of DoW test ranges and centers.

R&D Agreements - Partnership Intermediary Agreement (PIA)

Partnership intermediary organizations are the technology scouts and matchmakers for the DoW. They don't have funding attached to them, and cannot issue contracts or OTAs. But under the FAR DoW labs, program offices, and technology centers can work with a partnership intermediary organization—often a nonprofit, state/local government, or university entity—to help transfer technology and foster collaboration with startups. There are over 50 PIAs and they are executed decentrally by labs, warfare centers, and commands.

Keep in mind that getting a CRADA (Cooperative R&D Agreement) with a DoW lab accomplishes the same thing, or better. Appendix B lists and links to most PIAs.

R&D Agreements - Technology Investment Agreement (TIA)

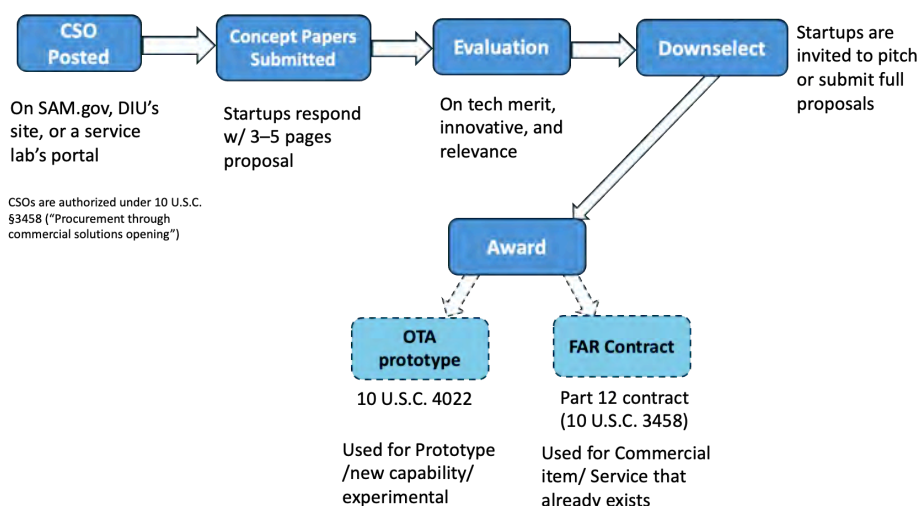
In the past, TIAs were used by DoW research offices (OUSD(R&E), DARPA, ONR, AFRL, ARL, NRL, DTRA, MDA) for pre-competitive, cost-shared R&D projects with startups. They're rare today because OTAs are more flexible, but still on the books and occasionally used

Commercial Solutions Opening (CSO)

CSOs are *not a standalone statute/contract type like OTAs or CRADAs*, but they are a *new competitive procedure*

authorized by DoW pioneered and used by the Defense Innovation Unit- DIU (and others) to quickly solicit and fund innovative commercial solutions.

CSOs at DIU are used to describe broad DoW problem statements, then down-select a small set of vendors who come



in and pitch and demo. DIU then offers quick contract awards - typically prototypes or research OTAs. Legally, CSOs solve the problem that the FAR doesn't allow rapid down-selects or "pitch day" style competitions easily. CSO's can be used to buy commercial products and services, R&D studies for commercial technology or to mature/transition commercial technology for use in the DoW.

CSOs are great for startups as they have negotiable payment milestones, terms and conditions, and intellectual property and data rights. And there is no requirement to adhere to government accounting standards. If the DoW partner who issued the CSO wants to deploy the prototype solution, OTA authority *allows them to issue a follow-on production contract without the need for further competition.*

Think of CSOs as an "open door" contracting method that looks more like how venture capital or industry scouting works than traditional DoW contracting.

Multiple Authority Announcement (MAA)

An MAA is a government contracting vehicle that combines multiple acquisition authorities under a single announcement. MAA's are specifically targeted toward disruptive technologies and rapid capability development from startups, and other innovative companies that might not engage with traditional government contracting processes.

MAAs eliminate the need for agencies to issue separate solicitations under different authorities, reducing administrative burden and accelerating the speed of purchasing. This allows agencies to use different types of contracts based on the nature of proposed work, including Broad Area Announcements (BAAs), Commercial Solutions Openings (CSOs), Other Transaction Agreements (OTAs), Procurement for Experimental Purposes (PEPs), and Cooperative Research and Development Agreements (CRADAs).

This means that as a company's research and product matures, a single MAA can span multiple solicitations supporting basic research and advanced development under one framework.

MAAs uses both Calls for Proposals, which use defined closing dates for specific government requirements, and Open Periods, which allow continuous submissions within specified timeframes, enabling offeror-generated ideas and concepts.

The Missile Defense Agency's implementation of MAA is called the Nimble Options for Buying Layered Effects (NOBLE). (Many of the MDA Golden Dome Solicitations will be via MAA.) The Air Force Research Lab (AFRL) calls their MAA program [the CHEERS program](#).

TRLs - Technology Readiness Levels

The Department of War (DoW) uses the Technology Readiness Level (TRL) scale to figure out how ready a company's technology is before they buy it. This framework was adapted from

[NASA's original 1-page TRL definitions](#), and is codified in the 89-page [DoW Technology Readiness Assessment \(TRA\) Guidebook, June 2023](#).

Think of TRLs as a way for the DoW to ask - Is this technology feasible? If it is feasible, will it work? And if it works can it be built and deployed at scale?



Questions about your startup product/service Technology Readiness Level (TRL) will come up in almost every conversation with a DoW lab, Program Office, Research Agency, etc. You should be clear where your product is now, and be able to articulate an evidence-based plan on how it will get to deployment.

TRL	Definition (Hardware)	TRL Example
1	Basic principles observed and reported	Scientific findings published
2	Technology concept and/or application formulated	Speculative use cases identified
3	Analytical and experimental proof-of-concept	Lab bench validation of concept
4	Component and/or breadboard validated in lab	Subsystem tested in lab
5	Component and/or breadboard validated in relevant environment	Subsystem in simulated field
6	System/subsystem prototype demo in relevant environment	Prototype test in field conditions
7	System prototype demo in operational environment	Prototype UAV flight test
8	Actual system completed and qualified through test and demo	Weapon subsystem qual'd via testing
9	Actual system proven through successful mission operations	Fielded system used in operations

HRLs - Human Readiness Levels

In August 2025, OSD (R&E) added long-needed - HRLs - Human Readiness Levels - to incorporate items like weight, layout and form, so they can be understood and designed in as early as possible during development.

The DoW has adopted the existing [ANSI/HFES 400-2021](#) standard.

A HRL primer is [here](#).² It will take awhile but eventually HRLs, like TRLs will start showing up in requirements.

Level	Technology Readiness Level	Human Readiness Level
9	Operational use of deliverable	System successfully used in operations across the operational envelope with systematic monitoring of human-system performance
8	Actual deliverable qualified through test and demonstration	Total human-system performance fully tested, validated, and approved in mission operations, using completed system hardware and software and representative users
7	Final development version of the deliverable demonstrated in operational environment	Human systems design fully tested and verified in operational environment with system hardware and software and representative users
6	Representative of the deliverable demonstrated in relevant environments	Human systems design fully matured as influenced by human performance analyses, metrics, prototyping, and high-fidelity simulations
5	Key elements demonstrated in relevant environments	User evaluation of prototypes in mission-relevant simulations completed to inform design
4	Key elements demonstrated in laboratory environment	Modeling, part-task testing, and trade studies of human systems design concepts completed
3	Concepts demonstrated analytically or experimentally	Requirements for supporting human performance established
2	Concept and application formulated	Human-focused concept of operations defined and human performance design principles established
1	Basic principles observed and reported	Relevant human capabilities, limitations, and basic human performance issues and risks identified

² <https://www.researchgate.net/publication/361248019>

Note to startups: While the DoW might be asking about your TRL or HRL, they don't tell the whole story.

What drives your company's readiness to deliver and the right path to sell, is not only the maturity of the problem-solution match and the ability of your company to build it, but the government willingness and capability to integrate it.

For example; If you have an immature technology and an immature company you need SBIR Phase I.

Or if you have a mature company, mature technology, but the government can't transition it, then the path might be to get a CRADA, educate the government, then jump to later stages in the diagram.

Or use the second diagram of going direct to the Warfighter.

Keep in mind the formula **TRL + HRL + ARL + IRL**.

TRL the Technology Readiness Level - is the state of your product.

HRL the Human Readiness Level - are the product human factors optimized for use.

ARL the Adoption Readiness Level - does the DoW know they have a problem, need a solution and have or can find a budget to pay for it

IRL the Investment Readiness Level - can you find the capital to fund your company.

SBIRs - Small Business Innovation Research Grants

As of March 20th 2026 i Congress reauthorized the SBIR Program.

One of the biggest changes is the addition of "Strategic Breakthrough Awards." These \$30M awards are modeled after STRATFI (Strategic Funding Increase) - a Air Force/AFWEX DoD-internal workaround to bridge the valley of death. The idea was to stack SBIR authorities for awards up to ~\$60M, for companies past Phase II but not yet at production scale. The new law codifies that model.

To qualify for a \$30M Strategic Breakthrough Award, a company must:

- Have at least one prior Phase II award
- Raise 100% matching funds from new private capital, new non-SBIR/STTR government funding, or a combination of both
- Demonstrate a technology validated by market research as an effective solution
- A commitment for inclusion in a Program Objective Memorandum, signed by a Program Acquisition Executive or higher
- Your technology must meet high-priority operational requirements with a clear path into acquisition
- At least 20% of your matching funds must come from new DoD funding outside of SBIR/STTR

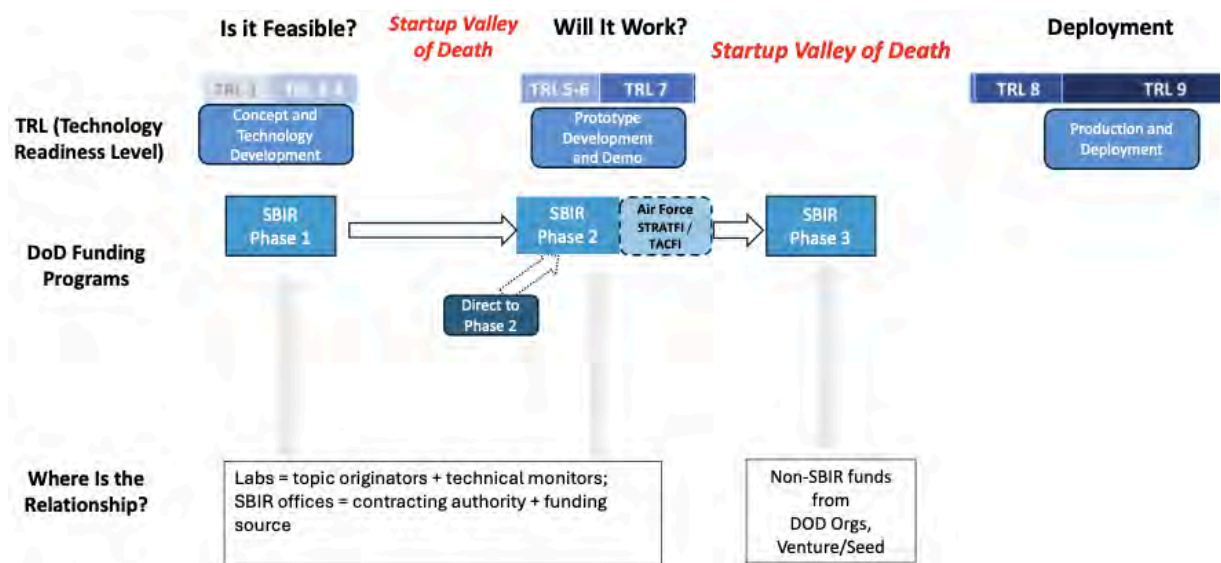
If you clear those bars you can get up to \$30M in SBA dollars over up to 48 months.

The matching requirement serves two purposes. It's a hook to encourage venture capital to participate in the government and defense innovation ecosystem. It also functions as a "demand signal" requirement. You have to prove that someone outside the SBIR program already believes in you enough to put money on the table. The DoD-specific commitments are interesting. Unlike many endorsement letters, a PAE-level signature is a program office saying they intend to buy. That is a signal for private investors, as it betrays DoW intent to transition.

The DoW Small Business Innovation Research - SBIR - program offers startups the equivalent of *pre-seed funding - in the form of grants - without taking any equity* (but in exchange for lots of paperwork.)

Each service, and the DoW, have their own SBIR/STTR office, and each is looking for different things and has different criteria. Look at the [DoW SBIR/STTR portal here](#)³ for all the active SBIR solicitations. Note that an SBIR process is not a “get big fast” process.

Applying for a DoW SBIR *makes sense for startups who don't have a working product or even a prototype*. You can get a SBIR Phase 1 grant to prove technical feasibility, and investigate potential military/commercial value, or to do a Concept Study. The deliverable of a Phase 1 SBIR could be limited prototyping and/or technology risk reduction.



If you get accepted to a SBIR, the money comes in two phases; Phase 1 \$50K - \$275K and if you apply and if you get accepted, a follow-on Phase 2 of up to \$2 million. Some companies with strong commercial traction can skip Phase I through [Direct-to-Phase II \(D2P2\)](#).

³ <https://www.dodsbirsttr.mil/submissions/login>

(One approach to selling to the DoW is to start with a small pilot R&D program and get it funded via SBIR grants and then hopefully turn that into an [OTA](#) and eventually a Program of Record (PoR).)

The goal of a DoW SBIR Phase II grant is to build and test a working prototype/functional demo in a relevant DoW environment. (Look for [Direct-to-Phase II \(D2P2\)](#) programs.)

One of the benefits of getting a DoW SBIR is that if a company had a SBIR Phase 1 or Phase 2, the government contractor can use DoW SBIR Phase III sole source authority to go directly to contract and forgo a more open competitive process.

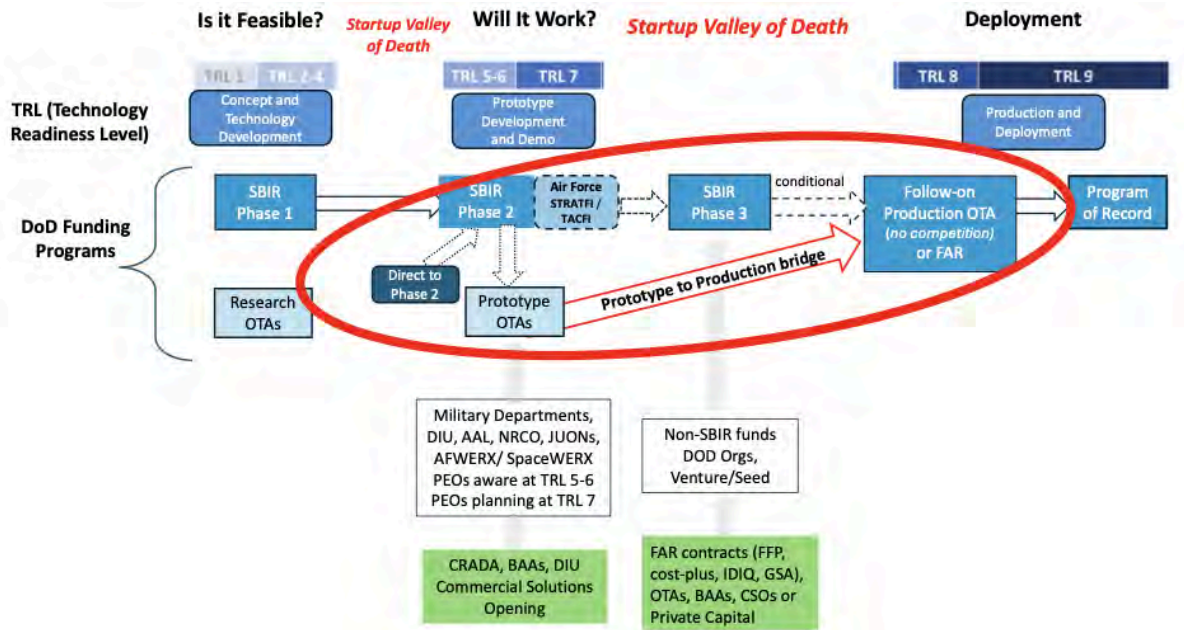
*What's Next After a SBIR Phase 2? - **TACFI/STRATFI***

The Air/Space Force both have bridge funds - Strategic Funding Increase ([STRATFI](#)) and Tactical Funding Increase (TACFI) Programs. Companies that have been awarded a Phase II contract within the last two years are eligible to apply. TACFI provides an additional \$375K-\$2M and STRATFI an additional \$3M-\$15M. An addition to March 2026 reauthorization is a "strategic breakthrough award" - a post-Phase II program that allows agencies to make awards up to \$30M

*What's Next After a SBIR Phase 2? - **SBIR Phase 3***

A SBIR Phase 3 contract needs to be funded with non-SBIR money (procurement, O&M, RDT&E, or even private/VC). The benefit of a SBIR Phase III is that it is eligible for a sole-source contract (no re-competition is required), it's unlimited in size and duration, and it preserves your company's SBIR data rights.

How to Get a SBIR Phase 3? Find a DoW Program Office / CPE (the customer with the real budget). They fund you with their dollars through a Phase III award. The contracting office can use an IDIQ, GSA schedule, or other vehicle. You don't have to wait on DoW money only. Phase III can be funded by Commercial sales, Prime contractor partnerships, Venture capital, other government agencies (NASA, DHS, DOE)



What's Next After a SBIR Phase 2? - **Prototype to Production OTA**

Many SBIR firms parlay their Phase II tech into a Prototype OTA (via a consortium like SpEC, IWRP, MCDC). If successful, this can roll into a non-competed Production OTA (like Phase III).

If you've been working the system, you would have found CPEs/PEOs /Program Managers as sponsors (the critical customer) to know about you when you are in the early-prototype stage in the lab, and sponsor you as your product was being tested in the field.

At this stage you may be ready to blend in commercial/VC funding to sustain development while DoW ramps.

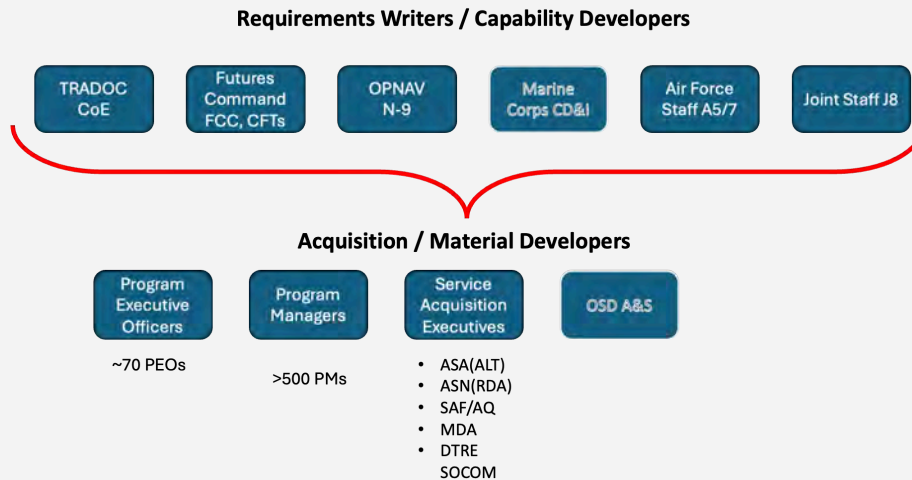
Search this document for the word "SBIR" and see Appendix B for details for how each of the Services SBIR programs differ.

Talk to the Requirements Writers (Capability Developers)

All the services have Requirements writers (Capability Developers) who draft Initial Capability Documents (ICDs), Capability Development Documents (CDDs), and Capability Production Documents (CPDs). These influence what the Services are looking to buy. Capability developers **do not** sit inside CPEs, but **they are now part of a larger PAE**.

DoW capability developers (requirements writers) mostly sit in Service requirements commands and centers (Navy OPNAV N9 (Warfare Systems) on the Navy staff, Marine Corps Combat Development & Integration (CD&I), in the Army's new Transformation and Training Command Future Command +, TRADOC Capability Development and Integration Directorate (CDID), Air Force Deputy Chief of Staff for Strategy, Integration and Requirements (AF/A5/7), and in Joint

Staff J8. They interface with Combatant Commands (CCMD J8/J9) for operational needs and with PEOs/PMs for acquisition.



Keep in mind:

- **Requirements Lane** (Capability Developers): Service Futures/Concepts commands, TRADOC CoEs, OPNAV N-9, Marine Corps CD&I, Air Force Staff A5/7, Joint Staff J8.
- **Acquisition Lane** (Materiel Developers): CPEs, PMs, Service Acquisition Executives (SAEs), OSD A&S.
- **Portfolio Acquisition Executives** - In theory will integrate Requirements and Acquisition Lanes

(The Materiel Developer is the organization or command responsible for acquisition, fielding, and sustainment of a materiel system. These are almost always the Capability Program Executive Offices (CPEs) / Program Executive Offices (PEOs) and their subordinate Program Managers (PMs) inside the Services and Defense Agencies.)

You need to talk to both the Requirements and Acquisition lanes inside PAEs.

Talk to the Service Front Doors, Service Labs, Future Concepts Centers and Forward Looking PAEs/CPEs PEOs

A few of the services have put their programs to onboard new companies in one place. For example, the Army has the Pathway for Innovation and Technology (PIT). The Army describes it as the “forward edge” of their acquisition arm - ASA(ALT). The PIT has programs and is designed to help companies to get into the defense industrial base. It’s the front door for startups wanting to partner with the Army.

The PIT includes the Army Applications Laboratory, Joint Innovation Outpost, the Global Tactical Edge Acquisition Directorate (G-TEAD) Marketplace, the [FUZE program](#) which includes ([Army SBIR/STTR](#) seed funding, [MANTECH](#) (Army Manufacturing Technology program), [TMI](#) (Tech Maturation Initiative) and [XTech](#) the Army’s scouting program.

As mentioned earlier, one (of many) possible startup go-to-market strategies is to look to the DoW Lab ecosystem for early CRADAs, research and prototype/R&D SBIR grants, and OTAs. The DoW lab ecosystem is big and varied – with 60+ labs, warfare centers, and engineering centers across the services. The ones that are practical “entry points”:

- Have discretionary budgets or rapid contracting authorities.
- Run open solicitations or challenges for nontraditional vendors.
- Have mandates to work with commercial tech and move quickly into prototyping.
- Write CRADAs with startups
- Have Partnership Intermediary Agreements (PIAs) and have given OTA authority to Consortia (both described further below.)

Below are a few places to start listed by services/commands. Details and links about these organizations can be found in the rest of this document.

Federal Labs

Army Labs

The [Army Application Laboratory](#) (AAL) - part of the Army Pathway for Innovation and Technology (PIT) - mission is to get the best technology to Army units faster by putting new solutions in the hands of Soldiers, either through pilot programs or with units for extended experimentation. AAL has [a list of open opportunities here](#).⁴ For startups, AAL’s ability to influence requirements documents is critical to help new technologies cross the valley of death. AAL’s strategic contribution is to grow the base of commercial technology firms solving Army tech problems. Read AAL’s [performance report](#) and then their website to get an idea of what they do.

[DEVCOM - the Army Combat Capabilities Development Command](#) focus is on Basic and Applied research for the Army, often 2- 5 years out creating prototypes, and technology seeded into Army programs. DEVCOM has 7 directorates.

DEVCOM also houses the [Army Research Laboratory](#) (ARL, Adelphi & Aberdeen). [See here⁵ for their engagement opportunities](#) with startups via: [Synchronized Persistent Army Experimentation \(SPARX\)](#), CRADAs (Cooperative Research and Development Agreements), SBIR/STTRs, Other Transaction Agreements (less frequent than the [Army Application Lab](#)) to onboard nontraditional vendors.

DEVCOM often partners via Consortia like [NAMC \(National Advanced Mobility Consortium\)](#). DEVCOM also offers Patent License Agreements (PLAs) to tech transfer their inventions out to companies.

The Army’s [xTechSearch](#) manages all the Army innovation prize competitions. [Open competitions here](#).⁶

⁴ <https://aal.mil/industry/#opportunities>

⁵ <https://devcom.army.mil/partner-with-us/>

⁶ <https://xtech.army.mil/competitions/>

[ERDCWERX](#) - offers collaboration and contracting with the Army Engineer Research and Development Center (ERDC) - part of the Army Corps of Engineers. [Events, tech challenges, and opportunities here](#).⁷

The [Army Software Factory](#), part of the Army's Future Command is an innovation program for active duty soldiers, SGT through SFC, CW2 and CW3, and 1LT through MAJ.

Navy Labs

[NavalX](#) - now part of the Navy Rapid Capabilities Office - offers startups opportunities via the [NavalX Tech Bridge Network](#), regional connections for Navy/Small Business collaboration.

The [Office of Naval Research](#) (ONR) publishes [a list of Broad Area Announcements](#) (BAAs) - notices of funding opportunities (NOFOs) and commercial solution openings (CSOs). Each announcement provides technical and contracting points of reference. See [here](#).⁸

The Navy Systems Commands - SYSCOMs - all use Broad Agency Announcements (BAAs) and SBIR/STTRs and OTAs. Look at [the Navy SBIR page here](#).⁹ A list of [all Navy SBIR/BAA announcements can be found here](#).¹⁰ Search this document for each system command small business office.

- [Naval Sea Systems Command](#) (NAVSEA)
- [Naval Air Systems Command](#) (NAVAIR)
- [Naval Information Warfare Systems Command](#) (NAVWAR)
- [Naval Facilities Engineering Systems Command](#) (NAVFAC)
- [Naval Supply Systems Command](#) (NAVSUP)
- [Marine Corps Systems Command](#) (MCSC)

The Navy has a [SBIR/STTR Transition Program](#) via the [Virtual Transition Marketplace](#)

Air Force Labs

[Air Force Research Laboratory \(AFRL\)](#) located at Wright-Patterson (Dayton, OH), Kirtland (Albuquerque), Rome (NY), and Eglin (FL). Currently, AFRL has the easiest "on-ramp" into DoW with high transition success.

AFRL's innovation programs include [AFWERX](#), the innovation arm of the Air Force and [AFWERX Ventures Program](#). Each year, AFWERX awards approximately 1,400 SBIR/STTR contracts worth more than \$1.4 billion. AFWERX developed the Strategic Funding Increase ([STRATEFI](#)) and Tactical Funding Increase (TACFI) Programs, a bridge for SBIR Phase II winners into programs of record.

⁷ <https://erdcwerx21.wpengine.com/category/event-tech-challenges/current>

⁸ <https://www.onr.navy.mil/work-with-us/funding-opportunities/announcements>

⁹ <https://www.navysbir.com/>

¹⁰ <https://www.navysbir.com/solicitations.htm>

See [here](#).¹¹ Small businesses that have been awarded a Phase II contract within the last two years are eligible to apply. TACFI provides an additional \$375K-\$2M and STRATFI an additional \$3M-\$15M. Note that the March 2026 proposed reauthorization allows for \$30M Phase II grants!

AFRLs sponsors the [Catalyst Accelerator](#) offering a variety of Air Force/Space Force topics in a three month semi-residential program in Colorado Springs or Ogden. They take no equity.

[STRIKEWERX](#) is an innovation arm of the Air Force Global Strike Command Office of the Chief Scientist.

Look at the [Air Force and Space Force Tech Connect](#) website for other Air Force opportunities.

If you're looking for ideas and/or technology, the [Air Force Tech Transfer Office](#) has thousands of technologies to transfer to industry. [See here](#).¹²

Space Force Labs

[SpaceWERX](#) (part of AFWERX) acts as the startup entry point for the U.S. Space Force. They run pitch days and commercial solution openings for space tech companies. [Space Ventures](#) is their investment arm via SBIR and STRATFI/TACFI. Space Spark connects end-users to lab engineers and acquirers to pursue novel solutions to their toughest challenges

The [Space Development Agency](#) also has a dedicated small business office.

SOCOM Labs

The Special Operations Command is one of the few combatant commands with its own innovation hub and acquisition authorities. [SOFWERX](#) looks for cutting-edge technologies that enhance the equipment and capabilities of U.S. special forces. Fast contract awards.

Defense Innovation Unit (DIU)

[The Defense Innovation Unit \(DIU\)](#) is the only DoW organization focused exclusively on fielding and scaling commercial technology across the U.S. military at commercial speeds. DIU has a \$billion budget for acquisition. DIU helps PAEs and program offices better adopt what industry has built.

DIU currently focuses on seven areas: Artificial Intelligence, Autonomy, Cyber and Telecom, Emerging Technology, Energy, Human Systems and Space, and has portfolio managers and staff.

DIU has OTA authority. For companies who already have off-the-shelf products, DIU's Commercial Solutions Openings (CSO) with 60-to-90 day timelines to contracts, posts their

¹¹ <https://afwerx.com/>

¹² <https://www.aft3.af.mil/The-Technology/>